Scaling UltraHigh-Resolution E3SM Land Model for Leadership-Class Supercomputers

Dali Wang[1], Chen Wang[2], Qinglei Cao[3], Jayesh Krishna[4], Danqing Wu[4], Weijian Zheng[4], Peter Schwartz[1], Fengming Yuan[1], Kathryn Mohror[2], and Peter Thornton[1]

[1]*Environmental Sciences Division*, *Oak Ridge National Laboratory*, Oak Ridge, US
[2]*Center for Applied Scientific Computing*, *Lawrence Livermore National Laboratory*, US
[3]*Department of Computer Science*, *Saint Louis University*, US
[4]*Computing, Environment, and Life Sciences Directorate*, *Argonne National Laboratory*, US

***Abstract*—This paper presents advancements in scaling the ultrahigh-resolution E3SM Land Model (uELM) for deployment on leadership-class supercomputers, addressing the increased demand for km-scale Earth system modeling. By focusing on km-scale ELM simulations, we enhance predictive capabilities for climate interactions, facilitating improved responses to climate change impacts on energy systems, agriculture, and water resources. Our approach leverages innovative software architecture optimizations, sophisticated data handling techniques, and advanced parallel processing, achieving strong scalability on two leadership supercomputers (2400 nodes (105,600 cores) on Summit, and 1200 nodes (76,800 cores) on Frontier). Results from extensive scalability assessments on the Summit and Frontier also demonstrate outstanding IO performance (close to 400 GB/s write throughput) and the model's ability to efficiently handle increasing computational demands. This study not only establishes uELM's capability for high-resolution simulations over vast geographical domains, but also sets a foundation for future Earth system modeling breakthroughs.**

***Index Terms*—Energy Exascale Earth System Model (E3SM), E3SM Land Model, Scalability, Parallel I/O**

I. Introduction

Earth System Models (ESMs) are crucial for studying the Earth's climate system and the interactions among its components, such as the atmosphere, land, and oceans. They predict both natural and human-induced climate changes, helping inform policies and enhance public awareness. Traditional ESMs use coarse grids, limiting their prediction accuracy for localized events.

The Energy Exascale Earth System Model (E3SM), funded by the U.S. Department of Energy (DOE), advances ESM capabilities by focusing on climate change and energy systems. It offers high-resolution options (e.g., 25 km grid spacing) that improve the representation of processes like oceanic eddies and storms. E3SM is now evolving towards ultra-high-resolution modeling utilizing exascale computing. The E3SM Land Model (ELM) plays a key role in this by detailing land processes, including energy, water exchanges, and carbon cycling. With the rising need for ultra-high-resolution models, optimizing ELM is critical for understanding climate impacts under various land use scenarios.

II. The Motivating Scaling Challenges

The push for ultra-high-resolution ELM (uELM) simulations is driven by the need for finer spatial and temporal resolutions in Earth system modeling, which current models fail to deliver. uELM aims to enhance predictive capabilities, allowing scientists to simulate complex interactions between land ecosystems and climate with remarkable detail. This high-fidelity simulation of extreme weather events and local features will improve our response to climate change impacts on energy systems, agriculture, and water management.

While uELM offers significant advancements in climate modeling, it poses considerable scaling challenges. A major concern is the computational intensity of uELM simulations (with more than half a million lines of code), where the increased number of grid cells leads to exponentially higher demands for computational power, storage, and memory. Effectively scaling uELM to utilize leadership-class supercomputers requires overcoming technical hurdles in resource allocation and load balancing.

uELM simulations produce an extraordinary volume of data, necessitating robust systems for processing, archiving, and analysis. To prevent data retrieval bottlenecks, we must develop scalable processing frameworks and storage solutions to ensure timely access to simulation outputs. Furthermore, integrating diverse datasets, such as soil properties, climate forcing, and topography, into a kilometer-scale framework requires innovative preprocessing and assimilation strategies. Addressing these integration challenges is essential to maintain the integrity of simulations and support scientific inquiry.

A. Related work

Most ESM simulations traditionally operate on coarse grids larger than 0.5×0.5 degrees [1]. The E3SM version 1 (E3SMv1) offers two resolutions: a standard 100 km grid and a high-resolution 25 km grid, facilitating more precise representation of ocean eddies and atmospheric phenomena [2]. E3SM is developing kilometer-scale models designed to exploit exascale computing capabilities, aimed at delivering detailed environmental representations.

Advancements in uELM simulation utilizing exascale computing for land simulations on continental and global scales have been reported [3], [4]. Some studies have utilized a simplified ELM model, incorporating a satellite phenology submodel and a reduced number of subgrid components, to investigate sub-grid topographic effects on land-atmosphere interactions in mountainous regions [5]. Recently, a kilometer-scale ELM model [6] was successfully executed on a su-

percomputer, demonstrating exceptional computational performance for the km-scale ELM simulation across North America. However, that effort also revealed significant challenges related to portability and data generation.

B. Innovations

The innovations of this study are the following.

1) Successfully ported the entire uELM model to two leadership-class computers with different system architecture and programming environments.
2) Demonstrated excellent end-to-end simulation scalability up to 2400 nodes (105,600 cores) and 1200 nodes (76,800 cores), respectively, on these two computers.
3) Archived outstanding IO performance on the Frontier and achieved 395 GB/s write performance with 1200 nodes.

III. Method

The uELM utilizes the Common Infrastructure for Modeling the Earth (CIME) for model configuration, construction, and execution on high-performance computers. We implement a data-driven ELM configuration with an active model using predefined atmospheric forcing data. This setup enables dynamic interactions among the energy, water, carbon, nitrogen, and phosphorus cycles, while also including soil organic matter decomposition and black carbon models[6].

To facilitate data integration, we developed specialized input files for custom grids at a 1 km x 1 km scale. The atmospheric and land models share the same computational domain for improved performance. We also employed MPI-based parallel processing to distribute workloads across multiple nodes, with each MPI process managing its subdomain for ELM calculations. Efficient output management is central to the uELM framework. The ELM generates time-averaged history outputs and restart files at user-defined intervals, supporting extensive simulations and data recovery. To enhance I/O performance, uELM uses the SCORPIO library[7], which optimizes data access by restructuring data across processes before interfacing with back-end libraries like PnetCDF, and ADIOS, alongside a parallel file system (GPFS, Lustre, and UnifyFS[8]).

This study utilized two supercomputers at Oak Ridge National Laboratory (ORNL): 1) Summit, which has 4,608 nodes with dual 22-core IBM POWER9 CPUs and six Nvidia Tesla GPUs. The experiments were developed using the ncvhp, alongside the NetCDF and PnetCDF libraries. 2) Frontier, featuring 9,408 nodes with 64-core AMD EPYC CPUs and 4 AMD MI250X GPUs, had experiments developed with the GCC and included NetCDF, PnetCDF, and ADIOS libraries.

IV. Scalability Evaluation

The base simulation, termed the AKSP case, is a uELM spin-up simulation over the Seward Peninsula in Alaska, encompassing 72,083 grid cells of 1 km x 1 km each [4]. As uELM operates implicitly without lateral transfers, we duplicated the AKSP domain to assess performance at varied scales. These experiments evaluated uELM's computational efficiency on high-performance systems. uELM simulations can extend for hundreds of years, typically generating monthly files and restart files every 10-20 years. For scalability tests, we constrained ELM to a brief (5-day) period with an hourly timestep, resulting in one history and one restart file that highlighted uELM's capabilities under limited resources. Table I lists the main input and output files for these experiments.

TABLE I: Size of major uELM input and output files (GB)

	AKSP	AKSPx10	AKSPx10x10	AKSPx10x10x3
Domain (I)	0.01	0.1	0.9	2.8
Forcing (I)	1.6	7.2	63.4	188.2
Surface (I)	7.1	50.3	497.7	1477.4
History (O)	0.48	4.81	47.91	134.18
Restart (O)	14.18	141.63	1,416.19	4,248.54

This study included two scalability assessments using the AKSP base simulation. The first focused on strong scalability, employing the AKSPx10x10x3 (721,624,900 grid cells) configuration across various node counts on Summit and Frontier. This setup evaluated computational performance improvements as processing units increased while maintaining a constant problem size. The strong scalability results indicated effective large-scale uELM simulations with efficient distributed workload and minimal overhead. The second assessment analyzed weak scalability, examining system performance as workload grew with additional nodes, ranging from 1 (AKSP) to 10 nodes (AKSPx10), 100 nodes (AKSPx10x10), and 300 nodes (AKSPx10x10x3). This approach aimed to assess the system's capacity to sustain performance as problem size increased with computational resources, defining the limits of uELM's efficiency in larger-scale scenarios.

A. Scaling experiments on Summit

Table II presents the scalability results of the land model (LND) component, excluding I/O operations. Table II (Top) shows the simulation times for LND in the strong scaling experiments, revealing a notable speedup of the LND component when scaling from 150 to 2,400 nodes; this speedup maintains 9.21 at 2,400 nodes with each core handling fewer than 250 grid cells. This decline can be attributed to overhead factors such as communication costs and synchronization delays. The weak scaling experiments (Table II (Bottom)) also exhibited remarkable scalability without the history and restart output as the LND execution times for the AKSP, AKSPx10, AKSPx10x10, and AKSPx10x10x3 cases ranges from 316.93 to 392.90 seconds. During the lengthy spin-up phase (>800 simulation years at hourly timestep), uELM typically produces history and restart files every 20 years. Consequently, the scaling outcomes of uELM without output in this configuration are significant.

TABLE II: Scaling result on Summit. Unit: second

Strong Scaling					
#node	150	300	600	1200	2400
Total Time	939.45	362.03	185.73	134.96	102.04
Speedup	1	2.42	5.08	6.96	9.21
Weak Scaling					
	AKSP	AKSPx10	AKSPx10x10	AKSPx10x10x3	
Total Time	292.9	358.08	364.35	362.03	

TABLE III: Weak scaling with I/O on Summit. Unit: second

	AKSP	AKSPx10	AKSPx10x10	AKSPx10x10x3
Total Time	317.50	637.66	559.47	1084.17
I/O Time	31.21	95.59	103.03	769.78
Write(GB/s)	1.6	3.3	16.4	8.6

Table III shows the weak scaling experiments with history and restart output. It has revealed additional issues concerning input/output (I/O) scalability with PnetCDF APIs. For the AKSP scenario, 31.21 seconds (10% of the total execution time) were spent on I/O operations, including a restart file of 14.18 GB with an average write bandwidth of 1.6 GB/s. The AKSPx10 and AKSPx10x10 scenarios allocated 15% and 18% of their time to IO operations, respectively. In the AKSPx10x10x3 scenario, IO activities accounted for over 70% of the LND execution time, where the uELM generated a restart file of 4,248.54 GB with an average write bandwidth of 8.6 GB/s. Therefore, in the following section, we utilize ADIOS on Frontier to improve I/O performance and address the limitations of PnetCDF.

B. Scaling experiments on Frontier

Table IV depicted LND weak scalability with IO time, including history and restart output. The computing time (total time minus I/O time) is consistently around 120 seconds for all four cases. In the PnetCDF experiments, the IO write bandwidth reached only 12.2 GB/s, leading to degraded weak scalability with more than 100 nodes. Conversely, experiments with ADIOS showed excellent scalability, achieving an IO write bandwidth of 186.3 GB/s with 300 nodes.

The strong scaling experiments (LND) presented interesting results (Table V): utilizing PnetCDF, the total execution time was 435.06 seconds with 600 nodes. Increasing the nodes to 1200 resulted in decreased computing and I/O times; however, initialization time rose, compensating for the decrease. The write bandwidth experienced a slight increase from 12.22 to 20.62 GB/s when transitioning from 300 to 1200 nodes.

TABLE IV: Weak scaling result on Frontier. Unit: second

	AKSP	AKSPx10	AKSPx10x10	AKSPx10x10x3
Total time[1]	144.01	167.62	279.40	516.01
I/O Time[1]	27.90	50.92	130.21	403.03
Write(GB/s)[1]	0.6	3.4	10.6	12.2
Total Time[2]	140.34	146.31	177.86	186.87
I/O Time[2]	25.77	30.01	59.21	64.42
Write(GB/s)[2]	0.7	6.5	38.3	186.3

[1] With PnetCDF, [2] With ADIOS.

TABLE V: Strong scaling result on Frontier. Unit: second

# nodes	150	300	600	1200
Total Time[1]	647.74	516.01	435.06	437.56
I/O Time[1]	406.52	403.03	365.91	286.92
Write(GB/s)[1]	11.97	12.22	14.06	20.62
Total Time[2]	-	186.87	158.92	238.90
I/O Time[2]	-	64.42	59.45	81.27
Write(GB/s)[2]	-	186.3	345.5	395.0

[1] With PnetCDF, [2] With ADIOS (- missing results, under investigation).

The uELM with ADIOS exhibited superior performance, with the total time reaching 158.92 seconds with 600 nodes. I/O time decreased to 59.45 seconds, accompanied by a write bandwidth of 345.5 GB/s. However, with 1200 nodes, the total time rose to 238.90 seconds, primarily due to an increase in I/O time (81.27 seconds), mainly from read operations (44.34 seconds), a significant increase from 29.68 seconds in the 600-node case. The average bandwidth indicated some improvement, reaching 395 GB/s. Another contributing factor to the increased time is the LND initialization, which is known to not scale well with a large number of nodes.

V. Conclusion

This study showcases the successful implementation of the complex uELM, composed of more than a half million lines of code, on two leadership-class supercomputers (Summit and Frontier). This effort enables the modeling of the Earth land component at kilometer scales, a task that was previously unachievable. The findings illustrate its robust scalability, supporting up to 2400 nodes (105,600 cores) on Summit and 1200 nodes (76,800 cores) on Frontier, along with exceptional I/O performance reaching nearly 400 GB/s). These achievements establish a foundation for future progress in high-resolution Earth system modeling.

We are working on scalability experiments with larger number of nodes on Frontier, and we are also porting the uELM code onto other supercomputers, including Perlmutter at the National Energy Research Scientific Computing Center. we will update the progress if our proposal is accepted for the May conference. We have also installed a new parallel file system UnityFS on Frontier, achieving 10.8 Gs/s write bandwidth in the AKSPx10 case with PnetCDF APIs using 10 nodes, approximately 4.23 times faster than the default Lustre file system. We will continue to improve uELM IO performance with UnifyFS on more nodes.

Furthermore, we have identified several future directions to enhance our research: 1) Investigate the uELM initialization scale issues; 2) Utilize node-local fast storage devices at larger scales to improve performance, and 3) Explore porting our applications to utilize GPUs for better computational efficiency [9], [3].

Acknowledgment

This research was supported as part of the Energy Exascale Earth System Model (E3SM) project funded by DOE's Office of Science, Office of Biological and Environmental Research. This project leverages Daymet data product, funded by NASA and the Oak Ridge National Laboratory (ORNL). This research used resources of the Oak Ridge Leadership Computing Facility and Computing ORNL, which are supported by the DOE's Office of Science under Contract No. DE-AC05-00OR22725. This work was also partially supported by the LLNL-LDRD Program under Project No. 23-ERD-053, performed under the auspices of the U.S. Department of Energy by Lawrence Livermore National Laboratory under Contract DE-AC52-07NA27344.

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